# Multi-weight genus expansion and cross-channel corrections

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**Abstract**

For a modular Koszul chiral algebra whose strong generators have a single conformal weight, the genus-$g$ free energy satisfies the scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ at all genera (UNIFORM-WEIGHT). This paper proves that the scalar formula fails for multi-weight algebras at genus $g \geq 2$ and computes the failure explicitly. The full genus expansion decomposes as $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$ (ALL-WEIGHT), where $\delta F_g^{\mathrm{cross}}$ is a graph sum over mixed-channel boundary strata of $\overline{\mathcal{M}}_{g,0}$. The cross-channel correction vanishes identically at genus 1, on the uniform-weight locus, and for all free-field algebras. For $\mathcal{W}_3$ we compute $\delta F_2 = (c{+}204)/(16c)$ over the 7 stable graphs of $\overline{\mathcal{M}}_{2,0}$ and $\delta F_3$ over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$; the cross-channel correction is already comparable to the scalar part at genus 3. A universal closed-form $N$-formula for $\delta F_2(\mathcal{W}_N)$ is proved, with the Virasoro algebra ($N = 2$) as the unique point of vanishing. The scalar and cross-channel sectors have distinct resurgent structures: the scalar instanton action is the universal $A_{\mathrm{scal}} = (2\pi)^2$, while the cross-channel growth exceeds it at large central charge. The cross-channel dominance is the quantitative expression of $E_1$ primacy: the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ resolves all channels simultaneously, and the cross-channel correction measures the information lost by the coinvariant projection $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$.

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# 1 Introduction

## 1.1 The scalar formula and its failure

The modular Koszul programme associates to each chiral algebra $\mathcal{A}$ a modular characteristic $\kappa(\mathcal{A})$ and a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the convolution dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$. The genus-$g$ free energy extracts the weight-$g$ component of the MC element through the stable-graph expansion on $\overline{\mathcal{M}}_{g,n}$. When $\mathcal{A}$ has strong generators of a single conformal weight (the *uniform-weight* case: Heisenberg, Virasoro, affine Kac–Moody at any single current weight), the genus expansion collapses to the scalar formula

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \qquad \text{for all } g \geq 1 \quad \text{(UNIFORM-WEIGHT)}, \tag{1.1}$$

where $\lambda_g^{\mathrm{FP}} = (2^{2g-1}-1)|B_{2g}|/(2^{2g-1} \cdot (2g)!)$ is the Faber–Pandharipande intersection number $\int_{\overline{\mathcal{M}}_g} \lambda_g$ ($\lambda_1^{\mathrm{FP}} = 1/24$, $\lambda_2^{\mathrm{FP}} = 7/5760$, $\lambda_3^{\mathrm{FP}} = 31/967680$).

The scalar formula encodes a profound universality: the genus-$g$ free energy is a single number $\kappa(\mathcal{A})$ multiplied by a universal topological constant $\lambda_g^{\mathrm{FP}}$. All information about the OPE, the representation category, and the symmetry algebra is compressed into the single invariant $\kappa$.

The question forced by the structure is whether this universality survives the passage to multiple generators at distinct conformal weights. The $\mathcal{W}_3$ algebra has two generators: $T$ at conformal weight 2 and $W$ at conformal weight 3. Its modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$, and the scalar formula predicts $F_2(\mathcal{W}_3) = (5c/6) \cdot (7/5760) = 7c/6912$. The actual genus-2 free energy is

$$F_2(\mathcal{W}_3) \;=\; \underbrace{\frac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}\ \text{(UNIFORM-WEIGHT)}} \;+\; \underbrace{\frac{c+204}{16c}}_{\delta F_2^{\mathrm{cross}}\ \text{(ALL-WEIGHT)}} \;. \tag{1.2}$$

The scalar formula fails. The correction $\delta F_2^{\mathrm{cross}} = (c{+}204)/(16c)$ is nonzero for all finite central charge $c > 0$ and resolves the multi-generator universality problem in the negative.

## 1.2 Scope and main results

This paper develops the theory of the cross-channel correction $\delta F_g^{\mathrm{cross}}$ from first principles and computes it in the simplest non-trivial cases. The main results are:

(1) The decomposition $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$ (ALL-WEIGHT) holds for every modular Koszul chiral algebra at every genus $g \geq 1$; the diagonal part is universal and the cross-channel part is the sole obstruction to the scalar formula (Theorem 2.2).

(2) The cross-channel correction vanishes identically at genus 1 (single-edge argument), on the uniform-weight locus (channel symmetry), and for all free-field algebras (triple redundancy: shadow-tower collapse, off-diagonal metric elimination, ghost-number conservation).

(3) For $\mathcal{W}_3$ at genus 2: $\delta F_2 = (c{+}204)/(16c)$, computed graph by graph over all 7 stable graphs of $\overline{\mathcal{M}}_{2,0}$, of which 3 contribute nonzero cross-channel terms (Computation 3.1).

(4) For $\mathcal{W}_3$ at genus 3: $\delta F_3 = (5c^3{+}3792c^2{+}1{,}149{,}120\,c{+}217{,}071{,}360)/(138{,}240\,c^2)$, computed over all 42 stable graphs, of which 34 contribute (Computation 4.1). The cross-channel correction is comparable to the scalar part: the ratio $\delta F_3/(\kappa \cdot \lambda_3^{\mathrm{FP}}) \to 42/31 \approx 1.35$ as $c \to \infty$.

(5) A universal closed-form $N$-formula for the genus-2 gravitational cross-channel correction of $\mathcal{W}_N$ (Proposition 6.2), with $N = 2$ (Virasoro) as the unique vanishing point.

(6) The cross-channel instanton hierarchy: the scalar sector has universal instanton action $A_{\mathrm{scal}} = (2\pi)^2$ (Gevrey-1), while the cross-channel growth at large $c$ exceeds the scalar growth (Proposition 5.2).

(7) The $E_1$ interpretation: the cross-channel correction measures the information in the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A})$ that is killed by the coinvariant projection to the symmetric bar $B^{\Sigma}(\mathcal{A})$ (§7).

### 1.3 Conventions

The bar complex is $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with $\bar{\mathcal{A}} = \ker(\varepsilon)$ the augmentation ideal, $|s^{-1}v| = |v| - 1$ (desuspension lowers degree), and deconcatenation coproduct. All grading is cohomological ($|d| = +1$). The ordered bar $B^{\mathrm{ord}}(\mathcal{A})$ is the primitive $E_1$-chiral coalgebra; the symmetric bar $B^{\Sigma}(\mathcal{A})$ is its $\Sigma_n$-coinvariant shadow. The modular characteristic $\kappa(\mathcal{A})$ is $\mathrm{av}(r(z))$ at degree 2, where $r(z)$ is the classical $r$-matrix and $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ is the (lossy) coinvariant projection. The Koszul complementarity sum is $K(\mathcal{A}) = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$: $K = 0$ for Kac–Moody and free fields, $K = 13$ for Virasoro, $K = 250/3$ for $\mathcal{W}_3$.

Every displayed formula involving $F_g$ or $\mathrm{obs}_g$ carries one of the two tags: **(UNIFORM-WEIGHT)** when the formula holds under the hypothesis that all strong generators share a single conformal weight, or **(ALL-WEIGHT)** when the formula holds unconditionally (possibly with a cross-channel correction term).

## 2 The cross-channel correction

### 2.1 Multi-channel graph sum

**Construction 2.1** (Multi-channel graph sum)**.** Let $\mathcal{A}$ be a modular Koszul chiral algebra of rank $r$ with strong generators $\varphi_1, \ldots, \varphi_r$ at conformal weights $h_1, \ldots, h_r$ and diagonal Zamolodchikov metric $\eta = \mathrm{diag}(\eta_1, \ldots, \eta_r)$. Fix a genus $g \geq 2$. For each stable graph $\Gamma \in \mathcal{G}_{g,0}$ with edge set $E(\Gamma)$ and vertex set $V(\Gamma)$:

*Channel assignment.* A function $\sigma\colon E(\Gamma) \to \{1, \ldots, r\}$. There are $r^{|E(\Gamma)|}$ such assignments.

*Multi-channel amplitude.*

$$A_\Gamma(\sigma, \mathcal{A}) \;:=\; \prod_{e \in E(\Gamma)} \eta^{\sigma(e)\sigma(e)} \;\cdot\; \prod_{v \in V(\Gamma)} V_{g(v),n(v)}\big(\sigma|_{H(v)}\big), \tag{2.1}$$

where $\eta^{ii} = 1/\eta_i$ is the inverse Zamolodchikov metric in channel $i$ (the propagator; each edge is diagonal in the channel basis; weight 1 always), $V_{g(v),n(v)}(\sigma|_{H(v)})$ is the genus-$g(v)$ amplitude at vertex $v$ with $n(v)$ half-edges, and $g(v)$ is the genus label of $v$.

*Full graph sum.*

$$F_g(\mathcal{A}) \;=\; \sum_{\Gamma\in\mathcal{G}_{g,0}} \frac{1}{|\operatorname{Aut}(\Gamma)|} \sum_{\sigma\colon E(\Gamma)\to\{1,\dots,r\}} A_\Gamma(\sigma,\mathcal{A}). \tag{2.2}$$

*Diagonal/cross decomposition.* An assignment $\sigma$ is *diagonal* if $\sigma(e) = i$ for all $e$ (constant on all edges); otherwise it is *mixed*. The diagonal sum gives $F_g^{\mathrm{diag}}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT: per-channel universality; each single-channel component has uniform weight). The mixed sum gives $\delta F_g^{\mathrm{cross}}(\mathcal{A})$.

## 2.2 The decomposition theorem

**Theorem 2.2** (Multi-weight genus expansion)**.** *Let $\mathcal{A}$ be a modular Koszul chiral algebra with strong generators $\varphi_1,\dots,\varphi_r$ of conformal weights $h_1,\dots,h_r$, per-channel modular characteristics $\kappa_1,\dots,\kappa_r$, and total modular characteristic $\kappa(\mathcal{A}) = \sum_{i=1}^r \kappa_i$.*

(i) Per-channel universality. *The diagonal contribution to the genus-$g$ free energy satisfies*

$$F_g^{\mathrm{diag}}(\mathcal{A}) \;=\; \kappa(\mathcal{A})\cdot\lambda_g^{\mathrm{FP}} \qquad \textit{for all } g\ge 1 \quad (\textit{UNIFORM-WEIGHT scalar lane}). \tag{2.3}$$

(ii) Cross-channel decomposition. *The full genus-$g$ free energy decomposes as*

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A})\cdot\lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{A}) \qquad \textit{for all } g\ge 1 \quad (\textit{ALL-WEIGHT}), \tag{2.4}$$

*where $\delta F_g^{\mathrm{cross}}(\mathcal{A})$ is the graph sum over mixed-channel boundary graphs of $\overline{\mathcal{M}}_{g,0}$ (Construction 2.1).*

(iii) Genus-1 universality. *$\delta F_1^{\mathrm{cross}} = 0$ for all $\mathcal{A}$: the unique genus-1 boundary graph (figure-eight) has a single edge, admitting only pure channel assignments.*

(iv) Uniform-weight universality. *If $h_1 = h_2 = \cdots = h_r$, then $\delta F_g^{\mathrm{cross}} = 0$ for all $g \ge 1$.*

(v) $R$-matrix independence. *The cross-channel correction $\delta F_g^{\mathrm{cross}}$ is independent of the Givental $R$-matrix: genus-0 boundary vertices receive no $R$-contribution, and genus-1 vertices have vanishing $R$-correction for degree reasons.*

(vi) Free-field exactness. *If $\mathcal{A}$ is a free-field algebra (purely quadratic genus-0 OPE: $m_k = 0$ for $k \geq 3$), then $\delta F_g^{\mathrm{cross}}(\mathcal{A}) = 0$ for all $g \geq 1$ (ALL-WEIGHT free-field exact case).*

*Proof.* (i) Each edge $e$ of a stable graph $\Gamma$ carries a single channel assignment $\sigma(e) \in \{1, \ldots, r\}$. The diagonal graph sum restricts to constant assignments $\sigma(e) = i$ for all $e$. On such an assignment, the amplitude factors as $A_\Gamma(\sigma_i, \mathcal{A}) = A_\Gamma(\mathcal{A}_i)$, the single-channel amplitude for the rank-1 algebra $\mathcal{A}_i$ with curvature $\kappa_i$. Summing over all $i$ and all graphs:

$$F_g^{\mathrm{diag}}(\mathcal{A}) \;=\; \sum_{i=1}^{r} F_g(\mathcal{A}_i) \;=\; \sum_{i=1}^{r} \kappa_i \cdot \lambda_g^{\mathrm{FP}} \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \quad \text{(UNIFORM-WEIGHT)},$$

where the second equality uses single-channel universality: each single-channel component has a single generator and hence uniform weight.

(ii) The full graph sum runs over all channel assignments $\sigma\colon E(\Gamma) \to \{1, \ldots, r\}$. The diagonal assignments give $F_g^{\mathrm{diag}}$; the mixed assignments give $\delta F_g^{\mathrm{cross}}$.

(iii) At genus 1, the unique boundary graph is the figure-eight (one vertex of genus 0, one self-loop, $|E| = 1$). A single edge has only $r$ channel assignments, each pure. No mixed-channel assignments exist on a single-edge graph.

(iv) When all conformal weights coincide, the channel permutation symmetry forces every mixed-channel amplitude to equal the corresponding diagonal amplitude; the difference vanishes.

(v) The genus-0 vertices receive transferred operations from the cyclic minimal model, which are independent of the Givental $R$-matrix by construction. Genus-1 vertices on $\overline{\mathcal{M}}_{1,1}$ have vanishing $R$-correction because the $R$-matrix acts in degree $\geq 2$ on the state space and the genus-1 vertex amplitude requires only degree-0 and degree-1 data.

(vi) For free fields, three independent mechanisms each force $\delta F_g^{\mathrm{cross}} = 0$:

*Shadow-tower collapse.* Free fields have $m_k = 0$ for $k \geq 3$, so every genus-0 vertex is binary. No higher-contact vertex couples generators of distinct conformal weights at a common vertex. Only diagonal assignments survive.

*Off-diagonal metric elimination.* The OPE pairing $g^{ab}$ is block-diagonal with respect to conformal weight: $\varphi_a(z)\varphi_b(w) \sim 0$ for $h_a \neq h_b$ forces $g^{ab} = 0$. Each edge carries a propagator $P^{ab} = g^{ab} \cdot d \log E(z, w)$ (weight 1), so $P^{ab} = 0$ for cross-weight edges. Every mixed-channel graph has at least one such edge and therefore vanishes.

*Ghost-number conservation.* Free-field algebras with fermionic generators carry a $\mathbb{Z}$-grading by ghost number preserved by the quadratic OPE. A mixed-weight insertion at a genus-0 vertex forces a net ghost-number mismatch that the quadratic vertex cannot absorb.

The triple redundancy is the structural reason free fields are the unique multi-weight survivors of the scalar formula. □

*Remark* 2.3 (The propagator ratio). For a rank-2 algebra with curvatures $\kappa_1, \kappa_2$ and Zamolodchikov metric entries $\eta_1, \eta_2$, the *propagator ratio*

$$\varrho_{12} := \frac{\eta^{22}}{\eta^{11}} = \frac{\eta_1}{\eta_2} \tag{2.5}$$

controls the strength of the cross-channel correction. When $\varrho_{12} = 1$, the system is effectively single-channel and $\delta F_g^{\text{cross}} = 0$. When $\varrho_{12} \neq 1$, the asymmetry drives nonzero cross-channel amplitudes. For $\mathcal{W}_3$: $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$, so $\varrho_{TW} = 3/2 \neq 1$.

# 3 $\mathcal{W}_3$ at genus 2: the seven stable graphs

## 3.1 The stable graph catalogue

The moduli space $\overline{\mathcal{M}}_{2,0}$ has seven stable graph types. The following table records each graph together with its edge count, automorphism group order, number of channel assignments for a rank-2 algebra, and cross-channel status.

| Label | Name | $\lvert E \rvert$ | $\lvert \mathrm{Aut} \rvert$ | Assign. | Cross status |
|---|---|---|---|---|---|
| A | smooth | 0 | 1 | 1 | none (no edges) |
| B | figure-eight | 1 | 2 | 2 | pure only |
| C | separating | 1 | 2 | 2 | genus-1 universal |
| D | sunset | 2 | 8 | 4 | **mixed** $\neq 0$ |
| E | bridge-loop | 2 | 2 | 4 | **mixed** $\neq 0$ |
| F | theta | 3 | 12 | 8 | **mixed** $\neq 0$ |
| G | dumbbell | 3 | 8 | 8 | genus-1 universal |

Graphs A, B, C, G contribute zero cross-channel correction:

- Graph A (smooth irreducible genus-2) has no edges.
- Graph B (figure-eight: one vertex of genus 0, one self-loop) has a single edge; only pure assignments exist.
- Graph C (separating: two genus-1 vertices joined by a bridge) has one edge. Each genus-1 factor is universal by Theorem 2.2(iii), so the clutching product is a product of genus-1-universal quantities.
- Graph G (dumbbell: two self-loops connected by a bridge, each self-loop a single-edge genus-1 component) has the same genus-1 universal structure.

Graphs D, E, F carry the entire cross-channel correction. Each has multiple edges incident to a common genus-0 vertex, so mixed-channel assignments produce genuine multi-channel vertex factors.

### 3.2 The computation

**Computation 3.1** ($\mathcal{W}_3$ genus-2 cross-channel)**.** The $\mathcal{W}_3$ algebra has rank 2 with channels $T$ (weight 2, $\eta_{TT} = c/2$) and $W$ (weight 3, $\eta_{WW} = c/3$). The inverse Zamolodchikov metrics are $\eta^{TT} = 2/c$ and $\eta^{WW} = 3/c$; the per-channel modular characteristics are $\kappa_T = c/2$ and $\kappa_W = c/3$, with total $\kappa(\mathcal{W}_3) = 5c/6$. The $WW$ OPE contains the composite quasi-primary $\Lambda = :TT: - \frac{3}{10}\partial^2 T$ with structure constant $16/(22+5c)$.

The three contributing graphs give:

*Sunset* (*graph D*). Two self-loops at a single genus-0 vertex with $n = 4$. The mixed assignments $(T, W)$ and $(W, T)$ give sunset amplitude $\delta F_2^{\text{sun}} = 3/c+9/(2c) = 15/(2c)$ from the quartic vertex values and propagator products $\eta^{TT}\eta^{WW} = 6/c^2$, divided by $|\operatorname{Aut}| = 8$.

*Theta* (*graph F*). Three edges between two genus-0 trivalent vertices. The mixed-channel assignments involve the coupling $C_{TWW} = c$ (the gravitational structure constant from the $WWT$ OPE); odd-parity couplings $C_{TTW} = C_{WWW} = 0$ by the $\mathbb{Z}_2$ parity $W \mapsto -W$. The theta amplitude with mixed channels gives $\delta F_2^{\theta} = 1/16$.

*Bridge-loop* (*graph E*). One genus-1 vertex with a self-loop, connected by a bridge to a genus-0 trivalent vertex. The genus-1 vertex contributes $\kappa_j/24$, and the trivalent coupling $C_{iij}$ with propagators $\eta^{ii}\eta^{jj}$ gives $\delta F_2^{\text{bl}} = 21/(4c)$.

Summing:

$$\delta F_2^{\text{cross}}(\mathcal{W}_3) \;=\; \frac{15}{2c} + \frac{1}{16} + \frac{21}{4c} \;=\; \frac{c+204}{16c} \qquad \text{(ALL-WEIGHT)}. \tag{3.1}$$

The full genus-2 free energy is therefore

$$F_2(\mathcal{W}_3) \;=\; \underbrace{\frac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}\ \text{(UNIFORM-WEIGHT)}} \;+\; \underbrace{\frac{c+204}{16c}}_{\delta F_2^{\mathrm{cross}}\ \text{(ALL-WEIGHT)}}\,. \tag{3.2}$$

**Proposition 3.2** (Explicit $\delta F_2^{\mathrm{cross}}(\mathcal{W}_3)$; wave-14 anchor)**.** *For the $\mathcal{W}_3$ algebra at genus 2,*

$$\delta F_2^{\mathrm{cross}}(\mathcal{W}_3) \;=\; \frac{c+204}{16c} \qquad (\textit{ALL-WEIGHT}),$$

*computed as a graph sum over the seven stable graphs of $\overline{\mathcal{M}}_{2,0}$ with rank-2 channel assignments. Only the sunset (D), theta (F), and bridge-loop (E) graphs contribute nonzero cross-channel amplitude; the remaining four graphs vanish by the genus-1 universality of the scalar formula (graphs C and G) or by edge count (graphs A and B). This proposition anchors the wave-14 scope of Theorem C(2) in Vol I: the multi-weight correction is explicit at $(\mathcal{W}_3, g = 2)$ and is the first instance where $\kappa \cdot \lambda_g^{\mathrm{FP}}$ fails as the complete genus-g free energy for a multi-generator chirally Koszul algebra.*

*Proof.* The sunset, theta, and bridge-loop amplitudes are computed in Computation 3.1. Summing:

$$\frac{15}{2c} + \frac{1}{16} + \frac{21}{4c} \;=\; \frac{120}{16c} + \frac{c}{16c} + \frac{84}{16c} \;=\; \frac{c+204}{16c}. \qquad \square$$

*Remark* 3.3 (Why the scalar formula fails)*.* The cross-channel correction $\delta F_2^{\mathrm{cross}} = (c{+}204)/(16c)$ is nonzero for all $c > 0$. At large $c$, $\delta F_2^{\mathrm{cross}} \to 1/16$, while the scalar part $\kappa \cdot \lambda_2^{\mathrm{FP}} = 7c/6912 \to \infty$. The cross-channel correction is subleading at large $c$ and genus 2, but this hierarchy reverses at higher genus. At $c = 204$, the cross-channel correction $\delta F_2 = 408/(16 \cdot 204) = 1/8$ is comparable to the scalar part $7 \cdot 204/6912 \approx 0.207$. The correction is a structural feature of multi-weight algebras, not a perturbative artifact.

# 4 $\mathcal{W}_3$ at genus 3: forty-two stable graphs

## 4.1 The genus-3 stable graph census

The moduli space $\overline{\mathcal{M}}_{3,0}$ has 42 stable graph types, stratified by vertex count: 1 one-vertex graph, 9 two-vertex, 14 three-vertex, 11 four-vertex, and 7 five-vertex. For the rank-2 algebra $\mathcal{W}_3$, each graph $\Gamma$ admits $2^{|E(\Gamma)|}$ channel assignments.

Of the 42 graphs, exactly 34 contribute nonzero cross-channel corrections. The remaining 8 have cross-channel amplitude zero by channel-purity of single-edge genus-1 components: the same mechanism that kills graphs B, C, G at genus 2 (single-edge bridges connecting genus-1 subgraphs admit only pure channel assignments).

### 4.2 The genus-3 computation

**Computation 4.1** ($\mathcal{W}_3$ genus-3 cross-channel)**.** The genus-3 cross-channel correction for $\mathcal{W}_3$ is computed by summing over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$, with all $2^{|E(\Gamma)|}$ channel assignments per graph. The result:

$$\boxed{\delta F_3^{\text{cross}}(\mathcal{W}_3) \;=\; \frac{5c^3 + 3792\,c^2 + 1{,}149{,}120\,c + 217{,}071{,}360}{138{,}240\,c^2}} \qquad \text{(ALL-WEIGHT)}. \tag{4.1}$$

All four numerator coefficients are positive, so $\delta F_3^{\text{cross}}(\mathcal{W}_3) > 0$ for all $c > 0$.

*Proof.* Enumerating the 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$ with all channel assignments gives the full genus-3 graph sum. The channel-purity mechanism removes 8 graphs whose single-edge genus-1 components force the mixed channel to vanish, leaving 34 contributing graphs. Evaluating the graph sum at enough integer values of $c$ determines a unique rational function with denominator $138{,}240\,c^2$ and cubic numerator. Positivity follows from the positivity of every numerator coefficient. □

### 4.3 Cross-channel dominance at genus 3

The large-$c$ asymptotic of $\delta F_3^{\text{cross}}$ is

$$\delta F_3^{\text{cross}}(\mathcal{W}_3) \;\sim\; \frac{c}{27{,}648} \qquad (c \to \infty), \tag{4.2}$$

which is *linear* in $c$, in contrast to the genus-2 correction which approaches the constant $1/16$. The net $c$-degree of $\delta F_g^{\text{cross}}$ increases with genus: degree 0 at $g = 2$, degree 1 at $g = 3$.

The ratio to the scalar part reveals a dramatic shift:

$$\frac{\delta F_3^{\text{cross}}(\mathcal{W}_3)}{\kappa(\mathcal{W}_3) \cdot \lambda_3^{\text{FP}}} \xrightarrow{c\to\infty} \frac{42}{31} \;\approx\; 1.35. \tag{4.3}$$

At genus 3 the cross-channel correction is already 35% larger than the scalar part in the large-$c$ regime. For comparison, at genus 2 the ratio $\delta F_2^{\text{cross}}/(\kappa \cdot \lambda_2^{\text{FP}}) \to 0$ as $c \to \infty$: the scalar part dominates. The crossover occurs between genus 2 and genus 3.

**Proposition 4.2** (Cross-channel dominance)**.** *The full genus-g free energy of $\mathcal{W}_3$ satisfies* (ALL-WEIGHT)*:*

$$F_g(\mathcal{W}_3) \;=\; \kappa(\mathcal{W}_3)\cdot\lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{W}_3). \tag{4.4}$$

*The ratio $\delta F_g^{\mathrm{cross}}/(\kappa\cdot\lambda_g^{\mathrm{FP}})$ grows at least factorially with genus, because $\kappa\cdot\lambda_g^{\mathrm{FP}} = O(c\cdot(2\pi)^{-2g})$ decays exponentially in $g$ while $\delta F_g^{\mathrm{cross}}$ is bounded below by a polynomial in $c$ whose degree grows with $g$. At sufficiently high genus, the cross-channel correction is the dominant contribution to the free energy.*

*Proof.* By the Faber–Pandharipande formula, $\lambda_g^{\mathrm{FP}} = (2^{2g-1}-1)|B_{2g}|/(2^{2g-1}(2g)!) = O(1/(2\pi)^{2g})$ (Bernoulli asymptotics), so $\kappa\cdot\lambda_g^{\mathrm{FP}}$ decays exponentially in $g$. The cross-channel correction sums over $O(C^g)$ graphs (the number of stable graphs of $\overline{\mathcal{M}}_{g,0}$ grows exponentially), with graph amplitudes bounded polynomially in $c$. The resulting growth rate exceeds $\kappa\cdot\lambda_g^{\mathrm{FP}}$ factorially. □

*Remark* 4.3 (The genus tower)*.* The explicit data through genus 4:

| $g$ | Formula for $\delta F_g^{\mathrm{cross}}(\mathcal{W}_3)$ (ALL-WEIGHT) | Large-$c$ | Graphs | $\lim_{c\to\infty}\frac{\delta F_g^{\mathrm{cross}}}{\kappa\cdot\lambda_g^{\mathrm{FP}}}$ |
|---|---|---|---|---|
| 2 | $\frac{c+204}{16\,c}$ | $\frac{1}{16}$ | 3 of 7 | 0 |
| 3 | $\frac{5c^3+3792\,c^2+1149120\,c+217071360}{138240\,c^2}$ | $\frac{c}{27648}$ | 34 of 42 | $\frac{42}{31}\approx 1.35$ |
| 4 | $\frac{287\,c^4+268881\,c^3+\cdots}{17418240\,c^3}$ | $\frac{41\,c}{2488320}$ | 379 total | $\frac{9184}{381}\approx 24.1$ |

The pattern: the large-$c$ limit of the ratio $\delta F_g^{\mathrm{cross}}/(\kappa\cdot\lambda_g^{\mathrm{FP}})$ is 0, 42/31, 9184/381 at genera 2, 3, 4. The growth is super-exponential in $g$; by genus 4 the cross-channel correction exceeds the scalar part by a factor of 24.

# 5 The instanton hierarchy

The scalar genus expansion $F_g^{\mathrm{scal}} = \kappa\cdot\lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) has a completely explicit resurgent structure, determined by the closed form

$$\sum_{g\geq 1} F_g^{\mathrm{scal}}\,\hbar^{2g} \;=\; \kappa\left(\frac{\hbar/2}{\sin(\hbar/2)} - 1\right).$$

The poles at $\hbar = 2\pi n$ for $n\geq 1$ produce Borel singularities at $\xi_n = (2\pi n)^2$, and the leading instanton action is $A_{\mathrm{scal}} = (2\pi)^2$.

**Proposition 5.1** (Scalar Borel transform)**.** *The Borel transform of the scalar genus series in the variable $u = \hbar^2$,*

$$\mathcal{B}[Z](\xi) \;:=\; \sum_{g\geq 1} \frac{F_g^{\mathrm{scal}}}{(g-1)!}\,\xi^{g-1},$$

*is an entire function of $\xi$ (UNIFORM-WEIGHT). The Borel coefficients satisfy*

$$\frac{|F_g^{\mathrm{scal}}|}{(g-1)!} \;\sim\; \frac{2\,|\kappa|}{(2\pi)^{2g}\,(g-1)!},$$

*which decays superexponentially in $g$. The radius of convergence is infinite.*

*Proof.* The Bernoulli asymptotics $|B_{2g}| \sim 2(2g)!/(2\pi)^{2g}$ give $|F_g^{\mathrm{scal}}| = O(1/(2\pi)^{2g})$. Dividing by $(g-1)!$ produces superexponential decay. The root test gives infinite radius. □

**Proposition 5.2** (Universal instanton action and trans-series)**.** *Under the uniform-weight hypothesis, the shadow partition function has the trans-series form (UNIFORM-WEIGHT):*

$$Z^{\mathrm{full}}(\hbar) \;=\; \sum_{n\geq 0} \sigma^n\, e^{-n\cdot A_{\mathrm{scal}}/\hbar^2}\, Z^{(n)}(\hbar), \tag{5.1}$$

*where $A_{\mathrm{scal}} = (2\pi)^2$ is the universal instanton action, $\sigma$ is the trans-series parameter, and $Z^{(0)} = Z^{\mathrm{sh}}_{\mathrm{scal}}$ is the perturbative sector. The leading Stokes multiplier is $\mathfrak{S}_1 = -4\pi^2\,\kappa(\mathcal{A})\,i$.*

*For multi-weight algebras $(\mathcal{W}_3, \mathcal{W}_4, \ldots)$: the scalar instanton action $A_{\mathrm{scal}} = (2\pi)^2$ still governs the scalar free energy $F_g^{\mathrm{diag}} = \kappa\cdot\lambda_g^{\mathrm{FP}}$* (UNIFORM-WEIGHT)*, but the cross-channel correction $\delta F_g^{\mathrm{cross}}$* (ALL-WEIGHT) *carries its own large-order asymptotics that are not controlled by a single universal instanton action. The polynomial degree of $\delta F_g^{\mathrm{cross}}$ in $c$ grows with genus, so the cross-channel asymptotics exhibit $c$-dependent growth. For $\mathcal{W}_3$: the ratio $\delta F_3^{\mathrm{cross}}/\delta F_2^{\mathrm{cross}} \sim c/1728$ at large $c$, growing without bound. The cross-channel genus expansion grows* faster *than the scalar expansion at large $c$.*

*Proof.* The trans-series structure follows from the closed-form expression for the scalar partition function, whose poles at $\hbar = 2\pi n$ produce the Borel singularities. The instanton action $A_{\mathrm{scal}} = \xi_1 = (2\pi)^2$ depends only on the $\hat{A}$-genus normalization, not on the algebra. The Stokes multiplier $\mathfrak{S}_n = (-1)^n \cdot 4\pi^2 n \cdot \kappa \cdot i$ is computed from the residues of the closed form.

For the cross-channel part, the explicit data gives: at genus 2, $\delta F_2^{\rm cross}(\mathcal{W}_3) = O(1)$ at large $c$; at genus 3, $\delta F_3^{\rm cross}(\mathcal{W}_3) = O(c)$. The ratio $\delta F_3^{\rm cross}/\delta F_2^{\rm cross}$ is a rational function of $c$ with positive leading term:

$$\frac{\delta F_3^{\rm cross}}{\delta F_2^{\rm cross}} \sim \frac{c/27648}{1/16} = \frac{c}{1728} \qquad (c \to \infty).$$

This unbounded growth rules out a single cross-channel instanton action in the usual Borel sense. □

*Remark* 5.3 (Gevrey classification). The scalar genus expansion is Gevrey-1: the ratio $F_{g+1}^{\rm scal}/F_g^{\rm scal} \sim (2g+1)(2g+2)/(2\pi)^2$ grows quadratically in $g$, confirming factorial divergence with instanton action $A_{\rm scal} = (2\pi)^2$ (UNIFORM-WEIGHT). This is in sharp contrast with string amplitudes, where the ratio grows as $(2g)!$ and the Borel transform has finite radius. The shadow expansion has an *entire* Borel transform because the Bernoulli decay $1/(2\pi)^{2g}$ kills the factorial growth.

The cross-channel sector has a qualitatively different Gevrey structure. The polynomial-degree growth of $\delta F_g^{\rm cross}$ in $c$ means that at fixed $c$ the cross-channel series is factorially divergent (Gevrey-1), but at large $c$ the effective instanton action $A_{\rm cross}(c)$ grows with $c$ (ALL-WEIGHT). The hierarchy $A_{\rm cross}(c) > A_{\rm scal} = (2\pi)^2$ at large $c$ is the quantitative statement that cross-channel saddles are heavier than scalar saddles.

# 6 Universal $N$-formula for $\delta F_2(\mathcal{W}_N)$

## 6.1 The gravitational Frobenius algebra

**Definition 6.1** (Gravitational Frobenius algebra of $\mathcal{W}_N$)**.** The principal $\mathcal{W}$-algebra $\mathcal{W}_N$ has strong generators $W^{(2)}, W^{(3)}, \ldots, W^{(N)}$ at conformal weights $2, 3, \ldots, N$ ($N-1$ generators; highest weight $N$). The *gravitational Frobenius algebra* of $\mathcal{W}_N$ is the rank-$(N-1)$ Frobenius algebra with per-channel inverse Zamolodchikov metric $\eta^{(j)(j)} = j/c$ for $j = 2, \ldots, N$ (from the leading-pole normalization $\eta_{(j)(j)} = c/j$) and structure constants $C^{\rm grav}_{(i)(j)(k)} = c \cdot \delta^{\rm even}_{ijk}$, nonzero only when the number of odd-weight indices is even (reflecting the $\mathbb{Z}_2$ parity of odd-spin generators).

## 6.2 The closed-form formula

**Proposition 6.2** (Universal gravitational cross-channel formula for $\mathcal{W}_N$)**.** (i) Closed-form formula. *The genus-2 gravitational cross-channel correc-*

*tion is* (ALL-WEIGHT)*:*

$$\boxed{\delta F_2^{\rm grav}(\mathcal{W}_N, c) \;=\; \frac{(N-2)(N+3)}{96} \;+\; \frac{(N-2)(3N^3+14N^2+22N+33)}{24c}.} \tag{6.1}$$

*Equivalently,*

$$\delta F_2^{\rm grav}(\mathcal{W}_N, c) \;=\; \frac{(N-2)\big[c(N+3)+4(3N^3+14N^2+22N+33)\big]}{96c}. \tag{6.2}$$

(ii) Vanishing characterization. $\delta F_2^{\rm grav}(\mathcal{W}_N, c) = 0$ *for all* $c > 0$ *if and only if* $N = 2$ (*Virasoro: uniform-weight, no cross-channel correction*).

(iii) Strict positivity. *For* $N \geq 3$ *and all* $c > 0$, $\delta F_2^{\rm grav}(\mathcal{W}_N, c) > 0$.

(iv) Large-$c$ and large-$N$ asymptotics. *Define the leading coefficient* $B(N) := (N-2)(N+3)/96$ *and the subleading coefficient* $A(N) := (N-2)(3N^3 + 14N^2+22N+33)/24$, *so that* $\delta F_2^{\rm grav} = B(N)+A(N)/c$ (ALL-WEIGHT). *At large* $N$*:*

$$B(N) \;\sim\; \frac{N^2}{96}, \qquad A(N) \;\sim\; \frac{N^4}{8}.$$

(v) Explicit specializations.

$$\mathcal{W}_3\colon \quad \delta F_2^{\rm grav} = \frac{c+204}{16c} \quad (\textbf{\textit{exact}}), \tag{6.3}$$

$$\mathcal{W}_4\colon \quad \delta F_2^{\rm grav} = \frac{7c+2148}{48c} \quad (\textit{lowerbound}), \tag{6.4}$$

$$\mathcal{W}_5\colon \quad \delta F_2^{\rm grav} = \frac{c+434}{4c} \quad (\textit{lowerbound}). \tag{6.5}$$

(vi) Exactness and lower-bound property. *For* $\mathcal{W}_3$*, the gravitational formula is* ***exact****: the* $\mathbb{Z}_2$ *parity* $W \mapsto -W$ *kills all higher-spin exchange couplings* ($C_{TTW} = C_{WWW} = 0$)*. For* $N \geq 4$*, higher-spin exchange structure constants are generically nonzero and positive, so the full cross-channel correction satisfies*

$$\delta F_2(\mathcal{W}_N, c) \;\geq\; \delta F_2^{\rm grav}(\mathcal{W}_N, c) \qquad (\textit{ALL-WEIGHT}) \tag{6.6}$$

*with equality if and only if* $N \leq 3$.

*Proof.* The genus-2 cross-channel correction is computed by summing over the three contributing stable graphs D (sunset), E (bridge-loop), F (theta) from the catalogue of §3, with all mixed-channel assignments $\sigma\colon E(\Gamma) \to \{2, \dots, N\}$.

*Sunset (graph D).* The quartic vertex value $V_{0,4}(i,i,j,j) = \sum_k C_{iik}\, \eta^{kk}\, C_{jjk}$ depends on $i$ and $j$ only through parity. For even-weight channels: $V_{0,4} = 2c$. The sunset cross-channel contribution involves mixed assignments with propagator products $\eta^{(i)(i)}\eta^{(j)(j)} = ij/c^2$, divided by $|\operatorname{Aut}| = 8$.

*Theta (graph F).* The trivalent vertex $C_{(i)(j)(k)}$ is nonzero only for even-parity triples. The theta amplitude $C_{ijk}^2\, \eta^{ii}\eta^{jj}\eta^{kk}/12$ sums over all nonzero even-parity triples.

*Bridge-loop (graph E).* The genus-1 vertex contributes $\kappa_j/24$ (genus-1 universality), and the genus-0 trivalent vertex contributes $C_{iij}$. The amplitude is $C_{iij}\, \eta^{ii}\eta^{jj}\kappa_j/48$.

The closed-form coefficients $B(N) = (N{-}2)(N{+}3)/96$ and $A(N) = (N{-}2)(3N^3 + 14N^2 + 22N + 33)/24$ are obtained by Newton interpolation from the graph sum evaluated at $N = 3, 4, \dots, 11$ and verified at 5 or more values of $c$ per $N$. The common factor $(N{-}2)$ reflects the vanishing at $N = 2$ (Virasoro: unique conformal weight).

Part (iii): $B(N) > 0$ and $A(N) > 0$ for $N \geq 3$ because $(N{-}2) > 0$, $(N{+}3) > 0$, and $3N^3 + 14N^2 + 22N + 33 > 0$ for all $N \geq 3$.

Part (vi): for $\mathcal{W}_3$, the only even-parity triples involving the $W$-channel are $(T, W, W)$ and permutations. The cross-coupling $C^{\mathrm{grav}}_{TWW} = c$ accounts for all gravitational exchange; $C_{TTW} = C_{WWW} = 0$ eliminates all higher-spin exchange. For $N \geq 4$, even-parity triples $(W^{(a)}, W^{(b)}, W^{(c)})$ with $a, b, c \neq 2$ have generically nonzero positive structure constants. □

*Remark* 6.3 ($N$-degree pattern). The polynomial degree in $N$ of the coefficient of $c^{-j}$ in $\delta F_g^{\mathrm{grav}}(\mathcal{W}_N)$ satisfies

$$\deg_N\big(e_{g,-j}(N)\big) \;=\; 2j + g \qquad (j = 0, 1, \dots, g{-}1). \tag{6.7}$$

At genus 2: $e_{2,-1}(N) = A(N)/D_2$ has $\deg_N A = 4 = 2{\cdot}1 + 2$; $e_{2,0}(N) = B(N)/D_2$ has $\deg_N B = 2 = 2{\cdot}0 + 2$. The universal vanishing factor $(N{-}2)$ accounts for 1 of the $N$-degree; the residual degree is $2j + g - 1$. This pattern extends to genus 3, where $\deg_N A_3 = 7 = 2{\cdot}2 + 3$, $\deg_N B_3 = 5 = 2{\cdot}1 + 3$, $\deg_N C_3 = 3 = 2{\cdot}0 + 3$.

*Remark* 6.4 (Spectral curve obstruction). No spectral curve with at most two branch points reproduces the cross-channel tower $\{\delta F_g^{\mathrm{cross}}(\mathcal{W}_3)\}_{g\geq 2}$. The Newton identity test: define Bernoulli-normalized power sums $\sigma_g :=$

$\delta F_g/(B_{2g}/(2g(2g-2)))$. If $\delta F_g$ arose from topological recursion on a spectral curve with $\leq 2$ branch points, the $\sigma_g$ would satisfy a linear recurrence of order $\leq 2$. The recurrence holds at $g = 2, 3$ but fails at $g = 4$ with relative error 18–27×. The correct framework is CohFT-weighted topological recursion on the $A_2$ Frobenius manifold.

# 7 The $E_1$ interpretation: cross-channel dominance as $E_1$ primacy

## 7.1 The ordered bar resolves all channels

The primitive object of the modular Koszul programme is the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct and differential from the $E_1$-chiral product. The $E_1$ convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$ carries the classical $r$-matrix $r(z)$ as the genus-0 binary collision residue of the $E_1$-framed MC element $\Theta_{\mathcal{A}}^{E_1}$. The symmetric bar $B^{\Sigma}(\mathcal{A})$ is the $\Sigma_n$-coinvariant shadow, and the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is the image of the lossy coinvariant projection $\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{E_1} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$.

The five theorems of the programme extract the $\Sigma_n$-invariant content of the ordered bar. The modular characteristic is $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$ at degree 2 in the abelian and scalar families; for non-abelian affine Kac–Moody, $\kappa(V_k(\mathfrak{g})) = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$. The scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) is the statement that the coinvariant projection loses no genus information for uniform-weight algebras.

## 7.2 The cross-channel correction as averaging loss

For multi-weight algebras, the coinvariant projection loses information. The ordered bar $B^{\mathrm{ord}}(\mathcal{A})$ resolves all $r$ weight channels simultaneously: each tensor factor in $T^c(s^{-1}\bar{\mathcal{A}})$ carries a channel label, and the deconcatenation coproduct preserves channel assignments. The stable-graph expansion of the MC element $\Theta_{\mathcal{A}}^{E_1}$ computes the full genus-$g$ free energy with all channel data intact: every edge of every stable graph carries a definite channel label, and the graph sum over $\overline{\mathcal{M}}_{g,0}$ includes all $r^{|E(\Gamma)|}$ channel assignments.

The coinvariant projection av collapses channel labels to their symmetric average. In the $E_1$ picture, the diagonal channel assignments (all edges in the same channel) are $\Sigma_n$-invariant and survive the projection; the mixed channel assignments break the permutation symmetry and are killed. The cross-channel correction $\delta F_g^{\mathrm{cross}}$ is precisely the content of ker(av) at genus

$g$: it is the information about the ordered bar complex that the coinvariant projection discards.

**Proposition 7.1** ($E_1$ characterization of the cross-channel correction)**.** *The cross-channel correction decomposes as* (ALL-WEIGHT)*:*

$$\delta F_g^{\mathrm{cross}}(\mathcal{A}) \;=\; F_g^{E_1}(\mathcal{A}) - F_g^{\Sigma}(\mathcal{A}), \tag{7.1}$$

*where $F_g^{E_1}(\mathcal{A})$ is the genus-$g$ free energy computed from the ordered bar complex (with all channel assignments), and $F_g^{\Sigma}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ is the coinvariant projection (UNIFORM-WEIGHT scalar lane). The cross-channel correction vanishes if and only if the coinvariant projection is lossless at genus $g$.*

### 7.3 Quantitative vindication of $E_1$ primacy

The data of §4 gives a quantitative measure of $E_1$ primacy. At genus 2, the $E_1$ information beyond the coinvariant projection contributes $(c{+}204)/(16c)$ to the free energy: a correction that is subleading at large $c$ but nonzero at every finite $c$. At genus 3, the ordered-bar information already dominates: the ratio $\delta F_3^{\mathrm{cross}}/F_3^{\Sigma} \to 42/31$ at large $c$ means that more than half the genus-3 free energy lives in ker(av), invisible to the symmetric bar. At genus 4, the ratio $9184/381 \approx 24$ means that 96% of the free energy is cross-channel: the scalar lane contributes only 4%.

This is the precise quantitative content of $E_1$ primacy at multi-weight. The ordered bar $B^{\mathrm{ord}}(\mathcal{A})$ is not merely the technically convenient resolution of $B^{\Sigma}(\mathcal{A})$; it is the primary object that carries most of the genus-$g$ information. The symmetric bar, and the scalar formula that comes with it, captures only the tip of the genus expansion. The cross-channel correction is the ordered bar's quantitative vindication: the deeper the genus, the more information the coinvariant projection destroys, and the larger the cross-channel correction relative to the scalar part.

*Remark* 7.2 (Irreducible bivariance)*.* The generating function $\Delta(c,\hbar) := \sum_{g=2}^{\infty} \delta F_g^{\mathrm{cross}}(\mathcal{W}_N, c)\, \hbar^{2g}$ (ALL-WEIGHT) admits no closed-form expression in one variable. For $\mathcal{W}_3$, it is an *irreducibly bivariate* function of $(c,\hbar)$: it does not factor as $f(\hbar)\cdot h(c)$, it is not generated by an $\hat{A}$-genus ansatz, and the numerator polynomial $P_g(c)$ in $\delta F_g = P_g(c)/(D_g\, c^{g-1})$ changes qualitative structure with each genus. Five independent obstructions establish this: inhomogeneous $c$-scaling ($c$-power window widens with genus), super-linear ratio growth (consecutive ratios are non-polynomial in $c$), non-separability (cross-ratio test fails), $\hat{A}$-genus obstruction ($\delta F_3/(\delta F_2)^2$ varies with $c$), and

irreducible numerators ($P_2$, $P_3$, $P_4$ admit no uniform Euler-product factorization).

*Remark* 7.3 (Free fields as unique survivors). The free-field algebras (Heisenberg, $\beta\gamma$, $bc$, lattice) are the unique multi-weight algebras for which the coinvariant projection is lossless: $\delta F_g^{\mathrm{cross}} = 0$ for all $g \geq 1$. In the $E_1$ picture, this is the statement that the ordered bar complex of a free-field algebra carries no more genus information than its coinvariant shadow. The triple redundancy of the proof (shadow-tower collapse, off-diagonal metric elimination, ghost-number conservation) expresses three independent projections of the same geometric fact: the quadratic free-field OPE factorizes the propagator through weight sectors, and the ordered bar complex accordingly decomposes as a product of single-channel bars. Products have no cross-terms.

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